



Hidden Markov Model Framework for Knowledge Correspondence Discovery and Semantic Interoperability across Ontologies

Josue Djimramadji^{1,2}, Lazarre Warda^{1,3*}, Kulemai Aymar Mamai¹, Jerome Mbainabeye², Guidedi Kaladzavi¹, Kolyang¹

¹ Laboratoire de Recherche en Informatique, The University of Maroua, 46 Maroua, Cameroon

² Department of Technology Sciences, University of Moundou, 206 Moundou, Chad

³ Department of Mathematics and Computer Science, Institut de Pedagogie d'Afrique Centrale, 121 Moundou, Chad

* Correspondence: Lazarre Warda (lwarda2015@gmail.com)

Received: 06-25-2025

Revised: 09-15-2025

Accepted: 09-23-2025

Citation: J. Djimramadji, L. Warda, K. A. Mamai, J. Mbainabeye, G. Kaladzavi, and Kolyang, "Hidden Markov Model framework for knowledge correspondence discovery and semantic interoperability across ontologies," *Int J. Knowl. Innov. Stud.*, vol. 3, no. 3, pp. 191–200, 2025. <https://doi.org/10.56578/ijkis030305>.



© 2025 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: Knowledge interoperability across heterogeneous systems poses a fundamental challenge to the predominantly knowledge-driven world, where semantic consistency directly affects data integration and knowledge exchange. Ontology alignment plays a pivotal role in facilitating semantic interoperability by identifying meaningful correspondences among concepts exemplified in different ontologies. This study investigated a probabilistic framework for ontology correspondence discovery using Hidden Markov Models (HMMs). In the proposed approach, each ontology was transformed into an HMM through Resource Description Framework (RDF) triple extraction using SPARQL (an RDF query language standardized by W3C) queries, where concepts were represented as hidden states and relationships as observation symbols. Both transition probability matrices and observation probability matrices were computed to characterize structural and semantic information embedded within the ontology. Concept vectors were then generated and compared by cosine similarity to identify potential correspondences among ontology concepts. Experimental evaluation was conducted using benchmark datasets from the Ontology Alignment Evaluation Initiative (OAEI). The results revealed that the proposed HMM-based framework achieved high precision and satisfactory recall, particularly for ontologies containing intricate relational structures. In addition, probabilistic modeling could effectively accommodate semantic variability and incomplete annotations without relying heavily on lexical resources or complex structural heuristics. The framework in this paper provides a superior mechanism for supporting semantic interoperability and contributes to knowledge integration across heterogeneous information.

Keywords: Interoperability; Ontology alignment; Semantic; Hidden Markov Models (HMMs); Knowledge

1 Introduction

With the rapid advancement of web technologies, modern applications acceleratingly prioritize knowledge over raw data to complement the breakthrough. A substantial portion of this knowledge was formalized through ontologies. An ontology O could be formally defined as a quadruple $(C, R, A, \text{ and } I)$, where C denotes a finite set of concepts serving as a proxy for entities within a specific domain. $R \subseteq C \times C$ is a finite set of relationships capturing interactions among concepts, A is a finite set of axioms specifying constraints governing concepts and relationships, and I is a finite set of individuals depicting concrete instances of these concepts [1].

Ontologies are commonly embodied by Resource Description Framework (RDF) triples [2], expressed as (Subject, Predicate, and Object), where Subject corresponds to a concept or individual. Object may be a concept, an axiom, or an individual whereas Predicate denotes the relationship linking the two. These triples could be queried using SPARQL (an RDF query language standardized by W3C) [3]. For the sake of clarity, the term "triple" refers to an RDF triple, and an ontology (O) can be formally characterized as:

$$O = \{(s, p, o) \mid s \in C \cup I, p \in R, o \in C \cup A \cup I\} \quad (1)$$

Ontologies are typically classified according to their levels of abstraction: Upper-level, core, and domain-specific ontologies. Prior to their construction, authors often employed techniques such as mapping, fusion, and reuse. Mapping aligns the concepts of one ontology with those of another; fusion merges multiple ontologies to create a new one whereas reuse imports some or all concepts from an existing ontology into a new ontology. Post-construction ontology alignment is performed to identify potential correspondences between concepts, usually through comparative analysis to determine a suitable target ontology.

Ontology alignment is fundamental for enhancing interoperability, knowledge reuse, data integration, and semantic search [4]. It enables collaboration across heterogeneous systems and domains, thus providing coherence and added value in environments where data is distributed and diverse. According to the Ontology Alignment Evaluation Initiative (OAEI) [5], ontology alignment not only promotes interoperability and knowledge reuse but also supports the evaluation and evolution of ontologies, thereby improving information systems.

The gradual integration of algorithmic techniques is targeting ontology manipulation alongside the advancement of artificial intelligence. Notably, Hidden Markov Models (HMMs) have demonstrated substantial effectiveness, as their inherent structural organization aligns naturally with the hierarchical and relational characteristics of ontologies [6].

Alignment approaches can be broadly classified into four categories: Those based on lexical resources [7–16], structural similarity [17–22], machine learning [23–35], and hybrid approaches [36–44]. Despite their effectiveness, these methods exhibit obvious limitations, including high computational complexity, incomplete information regarding ontology concepts, and challenges arising from diverse representation formats [45].

For ontologies that are rich in both concepts and relationships, a combination of multiple alignment strategies is often required. Within this context, the Hidden Markov Model (HMM) approach provides a probabilistic and sequential framework capable of capturing complex interdependencies among abstract concepts, while incorporating observable data. This approach is particularly well-suited to domains characterized by evolving and semantically intricate concepts.

In this study, we proposed a probabilistic ontology alignment approach based on Hidden Markov Model, in which concepts and relationships were modeled as hidden states and observation symbols [46]. Unlike prevailing lexical, structural, and conventional machine learning methods for ontology alignment, the proposed framework captured sequential and relational dependencies within ontologies, hence improving robustness to heterogeneous and incomplete representations. A similarity assessment mechanism grounded in HMM parameter estimation was proposed and experimentally validated.

The remainder of this manuscript is organized as follows: Section 2 introduces the formal definition of HMMs and reviews relevant works within the aforementioned categories of alignment approaches. Section 3 details the proposed methodology by outlining each procedural step. Section 4 details the experimental setup, results, and discussion, while Section 5 concludes the paper and highlights potential directions for future research.

2 State of Art

2.1 Definition

HMM is a stochastic model $\lambda = \{N, M, A, B, \pi\}$ [46] where:

- $N \in \mathbb{N}$, finite, is the number of HMM states;
- $M \in \mathbb{N}$, finite, is the number of HMM symbols;
- $A = [a_{ij}]$, $1 \leq i, j \leq N$, is the states transition probabilities distribution matrix of the model, where a_{ij} is the probability of transitioning from state i to state j and $\forall i \in [1...N]$, $\sum_{j=1}^N a_{ij} = 1$;
- $B = [b_{jk}]$, $1 \leq j \leq N$, $1 \leq k \leq M$, is the observation symbols probabilities distribution matrix of the model, b_{jk} is the observation probability of symbol k at the state j and $\forall j \in [1...N]$, $\sum_{k=1}^M b_{jk} = 1$;
- $\pi = [\pi_i]$, $1 \leq i \leq N$, is the initial states probabilities distribution of the model, and $\sum_{i=1}^N \pi_i = 1$.

The authors associated ontologies with HMM for building or populated ontologies and constructed systems based on knowledge; in [46], an approach was proposed to transform ontology into HMM.

2.2 Related Works

Ontology alignment constitutes a challenging task focused on establishing semantic correspondences between concepts across distinct ontologies. To address the inherent complexities of this process, a variety of methodologies has been developed, each offering particular advantage depending on the specific context. These methodologies can be broadly classified into four categories: Approaches based on lexical resources, methods relying on structural similarity, machine learning-driven techniques, and hybrid strategies.

2.2.1 Lexical resources-based approaches

These methods leverage lexical resources, such as WordNet or BabelNet, to analyze semantic correspondences between terms across ontologies. The underlying methodology involves comparing the terms or descriptions

associated with concepts using these lexical databases. Semantic relationships, including synonyms and hypernyms, are exploited to assess whether two distinct concepts refer to the same entity. While such approaches are effective in domains with a controlled vocabulary, their applicability is constrained by the coverage and completeness of the available lexical resources. Notable contributions in this category span a wide range of strategies. Alhassan et al. [7] extended an existing alignment system with a lexical database to improve the resolution of terminological ambiguities between heterogeneous ontologies. Similarly, Xue and Wu [8] optimized biomedical ontology alignment in lexical vector space, to illustrate that vector-based term representations enhanced alignment precision in specialized domains. In a related vein, Real et al. [9] leveraged domain-specific lexicons and grammar formalisms to constrain the matching process and reduce alignment errors, while Patel and Jain [10] combined terminological analysis with conceptual similarity measures to propose a computationally tractable alignment discovery method. Building on these foundations, Ibrahim et al. [11] responded to multilingual ontology matching and assessment, while Peng et al. [12] investigated ontology matching using textual class descriptions. More recently, Schneider et al. [13] have advocated Natural Language Focused Ontology Alignment (NLFOA), a system that interpreted concept descriptions using Natural Language Processing (NLP) techniques to extend applicability to loosely formalized ontologies. Along the same line, Guo et al. [14] advanced EventOA, a benchmark for event ontology alignment grounded in FrameNet and Wikidata, thus providing a rigorous evaluation framework for event-oriented alignment systems. Complementing these efforts, Silva et al. [15] boosted ontology matching through lexical and syntactic standardization when customized lexical analyzers were incorporated. Last but not least, Sharma et al. [16] proposed LSMatch, a large-scale ontology matching system designed in anticipation of high-volume alignment scenarios. Further related works are surveyed in current academic literature by [47].

2.2.2 Structural similarity-based approaches

Structural approaches emphasize the organization of concepts within ontologies. In these methods, ontologies are typically exemplified as graphs, where nodes correspond to concepts and edges denote the relationships between them. By comparing the graph structures, similarities between concepts can be inferred based on their hierarchical positions or their connections to other concepts. Such approaches are particularly well-suited to formal ontologies; however, they may encounter difficulties when significant structural divergences exist between the ontologies. Notable contributions in this category are ordered as follows: Chu et al. [17] optimized ontology alignment in vector space by exploiting the geometric properties of concept representations to refine correspondence identification. Subsequently, Xiang et al. [18] proposed Ontology-Guided Entity Alignment (OntoEA) via joint knowledge graph embedding. In a complementary effort, Guo et al. [19] investigated the improvement of knowledge graph embeddings for entity alignment. Furthermore, Hnatkowska et al. [20] advocated a fuzzy logic framework for ontology instance alignment, in order to offer a principled mechanism to handle uncertainty in partial or imprecise concept correspondences. Building upon structural embeddings, Hao et al. [21] combined semantic and structural representations for ontology alignment, while Şentürk and Aytac [22] suggested a graph-based ontology matching framework, particularly catered for formal ontologies with complex relational structures.

2.2.3 Machine learning-based approaches

Machine learning-based approaches employ algorithmic models to facilitate ontology alignment. In these methods, a model is trained on annotated datasets to predict correspondences between concepts. Techniques such as neural networks and embedding models, including Bidirectional Encoder Representations from Transformers (BERT), are utilized to analyze concepts and their interrelations. This paradigm has demonstrated remarkable performance, particularly when substantial and high-quality training data are available. However, it could entail considerable computational demands. Notable contributions related to machine learning in ontology alignment originated from a body of literature [31–35]. The machine learning category encompasses a broad spectrum of techniques that merit individual acknowledgment. Iyer et al. [23] presented VeeAlign, a supervised deep learning approach that learnt concept correspondences from annotated data, hence achieving competitive performance on standard alignment benchmarks. In contrast, Chakraborty et al. [24] introduced OntoConnect, an unsupervised recursive neural network-based system that aligned ontologies without labeled data by exploiting the graph hierarchical structure. Establishing upon evolutionary strategies, Xue et al. [25] applied linkage learning on entity correspondences to frame alignment as a search over the space of candidate mappings. Furthermore, Chen et al. [26] augmented ontology alignment by combining semantic embeddings with distant supervision, to mitigate dependence on manually annotated training data. In the biomedical domain specifically, Hao et al. [27] brought medical data to the field of ontology (MedTO), a hybrid graph neural network system for matching medical data to ontologies. Shifting toward generative approaches, Xue and Huang [28] applied generative adversarial networks to alignment optimization, hence improving robustness against noisy concept descriptions. With respect to annotation cost reduction, VersaMatch, a weakly supervised system upheld by Fürst et al. [29], demonstrated that automatically labeled data could yield quality comparable to fully supervised approaches. Subsequently, Norouzi et al. [30] explored conversational ontology alignment with extensive language models, to formulate an interactive paradigm

guided by natural language dialogue. Cotovio et al. [31] developed Matcha-DL, a supervised alignment tool, while Touati and Kemmar [32] applied deep reinforcement learning to the ontology matching problem. Sangeetha and Vidhyapriya [33] conceptualized a MapReduce-based approach for biomedical ontology alignment, and Gupta et al. [34] converged to suggest an ontology alignment method based on machine learning for the integration of patient health data. To offer a fresh perspective, Giglou et al. [35] leveraged large language models for ontology matching through their LLMs4OM framework.

2.2.4 Hybrid approaches

Hybrid approaches integrate multiple techniques to enhance the accuracy and robustness of ontology alignments. Typically, these methods may initiate the process using a lexical strategy to identify potential correspondences, which are then refined through structural comparisons or machine learning models. By combining the strengths of diverse methodologies, hybrid approaches are particularly effective in solving the inherent limitations of individual techniques. High-performing tools such as LogMap and Agreement Maker Light (AML) exemplify hybrid methods and frequently participate in initiatives like the OAEI. The OAEI is an annual benchmark designed to evaluate the performance of ontology alignment systems across various domains by providing a standardized framework for comparison. Notable contributions in the category of hybrid approaches encompass the works of Javed et al. [40], Xue [41], Amini et al. [44], Hartendorp et al. [43], and Snijder et al. [42]. The following contributions further elucidate the diversity of hybrid strategies in ascending order of reference. Bento et al. [36] applied convolutional neural networks to ontology matching, via integrating lexical and graph-topological features within a unified deep learning architecture. Along the same line, Neutel and de Boer [37] demonstrated that BERT-based contextual embeddings effectively encoded ontological concept labels, in a way outperforming conventional string-based similarity measures. Extending this line of work, He et al. [38] applied domain-adaptive BERT pre-training to biomedical ontology alignment. They substantially enhanced accuracy in medically specialized knowledge bases. He et al. [39], building further on transformer-based methods, presented BERTMap, which combined BERT embeddings with logical reasoning modules to achieve robust and interpretable ontology alignment. Subsequently, Javed et al. [40] proposed a deep ontology alignment approach using BERT INT applied to industrial IoT systems. In an attempt to tackle complex ontology alignment for autonomous systems, Xue [41] adopted a compact co-evolutionary optimization algorithm, while Snijder et al. [42] combined large language models with domain knowledge to advance ontology alignment in the labor market. Furthermore, Hartendorp et al. [43] initiated a self-alignment BERT model for biomedical entity linking in Dutch. Complex ontology alignment using large language models was explored in the study of Amini et al. [44], in order to extend the applicability of transformer-based methods to structurally intricate ontologies.

Ontology alignment, a continuously evolving field, encompasses a diverse range of techniques, from lexical resources to machine learning-based methods. Among these, hybrid approaches that integrate multiple strategies tend to yield the most robust results, owing to their flexibility and adaptability across different domains. The OAEI serving as a platform for benchmarking alignment systems, plays a pivotal role in the assessment and continuous improvement of these methodologies by providing standardized datasets and evaluation frameworks. Through the combination of these approaches, ontology alignment continues to advance toward increasingly precise and adaptable solutions to meet the demands of modern semantic systems [48].

The application of HMMs in ontology integration characterizes an apparent advancement, as HMMs could capture complex interrelationships among concepts, while preserving the underlying data structure. Several existing studies, as outlined in [46], have demonstrated the effectiveness of associating ontologies with HMMs, particularly in handling semantic variability and advancing integration tasks. Unlike conventional approaches, HMMs learn directly from input data without relying on pre-established lexical resources; therefore, they were more appropriate for domains that are sparsely documented. Their probabilistic framework enables flexible modeling of interactions among concepts to facilitate precise term-to-term alignments. Furthermore, HMMs exhibit advantages over traditional machine learning techniques by effectively tackling the uncertainty inherent in semantic relationships. Besides, they could provide a more nuanced understanding of contextual dependencies. This sophisticated approach could leverage both supervised and unsupervised learning paradigms and is adaptable to a variety of datasets, including those that are partially annotated.

3 Proposal Approach

All steps of this proposal are illustrated in Figure 1 and described with Algorithm 1.

Algorithm 1: HMM-based ontology concept correspondence

Input: Source ontology $O_s = (C_s, R_s, A_s, I_s)$, Target ontology $O_t = (C_t, R_t, A_t, I_t)$

Output: Set of correspondences $\mathcal{M} = \{(c_s, c_t, \sigma(c_s, c_t))\}$

$\mathcal{M} \leftarrow \emptyset$

Step 1: HMM-based ontology learning

- Extract RDF triple sets T_s and T_t from O_s and O_t ;
- Construct concept label set $L = C_s \cup C_t$ and $N = |L|$;
- Label triples with integers: T_s^l and T_t^l ;
- Generate HMM vector representations:

$$V_s = \{v_{s_i} \in \mathbb{R}^d \mid c_{s_i} \in C_s\}, \quad V_t = \{v_{t_j} \in \mathbb{R}^d \mid c_{t_j} \in C_t\}$$

Step 2: Vector similarity computation

For each $v_{s_i} \in V_s$ do:

- Compute distances $d_{ij} = \|v_{s_i} - v_{t_j}\|$ for all $v_{t_j} \in V_t$;
- Determine optimal match: $j^* = \arg \min_j d_{ij}$;
- Compute similarity score: $\sigma(c_s, c_t) \leftarrow f(d_{ij^*})$;
- Add correspondence to $\mathcal{M}$: $\mathcal{M} \leftarrow \mathcal{M} \cup \{(c_{s_i}, c_{t_{j^*}}, \sigma(c_{s_i}, c_{t_{j^*}}))\}$.

Return $\mathcal{M}$.

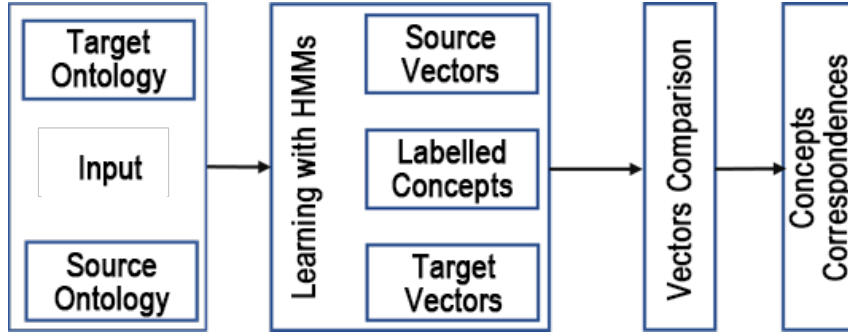


Figure 1. Hidden Markov Model (HMM)-based Alignment Process

3.1 Input

Ontology alignment constitutes a critical step in the ontology construction process [4]. During this phase, ontology developers seek existing definitions in other ontologies to minimize redundancy and promote reuse. Consequently, the source ontology, which may be the one currently under development, serves as the input for the alignment procedure, as illustrated in Figure 1. The set of target ontologies is selected from those emerging from the same domain, yet the number of these target ontologies is not inherently limited. It should be noted that a single ontology may be aligned with multiple target target ontologies simultaneously.

3.2 Learning with Hidden Markov Models

The learning phase with HMMs aims to embody the structural and semantic characteristics of ontologies with probabilistic modeling. Each ontology described in Section 3.1 was transformed into an HMM following the methodology proposed by Warda et al. [46]. In this framework, HMM states correspond to ontology concepts, while observation symbols correspond to ontology relationships.

Initially, a series of SPARQL queries were executed on an ontology, excluding axioms, and the results were signified as RDF triples. These triples were subsequently labeled and employed to initialize the HMM parameters, including the state transition probability matrix A , the observation probability matrix B , and the initial state distribution vector π , in accordance with the formulations presented in [46]. The labeling process involved replacing each entity identified by its IRI (Internationalized Resource Identifier, a chain of characters used to identify an identity within ontology) with a numeric identifier to facilitate subsequent computational manipulations (comparing numbers is easy than comparing too long chains of characters).

Let $T = \{(x, y, z) \mid x, z \in C', y \in R'\}$ denote the set of labeled triples, where C' and R' are the sets of labeled concepts and relations, respectively. The HMM parameters were initialized based on Eqs. (2)–(4).

$$A_{ij} = \frac{|\{(a, b, c) \mid (a, b, c) \in T, a = i, c = j\}|}{|\{(d, e, f) \mid (d, e, f) \in T, d = i\}|} \quad (2)$$

The element A_{ij} represents the probability that the concept labeled i serves as the subject while the concept labeled j serves as the object in a triple.

$$B_{ij} = \frac{|\{(a, b, c) \mid (a, b, c) \in T, a = i, b = j\}|}{|\{(d, e, f) \mid (d, e, f) \in T, d = i\}|} \quad (3)$$

where, B_{ij} denotes the probability that the concept labeled i is the subject and the relation labeled j is the predicate of a triple.

$$\pi_i = \frac{|\{(a, b, c) \mid (a, b, c) \in T, a = i\}|}{N} \quad (4)$$

π_i represents the probability that the concept labeled i appears as the subject of a triple.

3.3 Vectors Comparison

Once the HMM parameters were computed using Eqs. (2)–(4), the output of this step was generated, including source and target vectors (associated with the corresponding ontologies) as well as labeled concepts. The vectors correspond to the rows of the state transition probability matrix A of the HMM, where each row A_i , $i \in [1, N]$ represents the probability of a relationship between the concept labeled i and all other concepts as well as corresponding to v_{s_i} or v_{t_j} .

Vector comparison involved evaluating the similarity between each source vector and all target vectors. During this process, the index of the concept was retained to identify the one yielding the highest similarity. Various similarity measures could be employed for this comparison. Labeled concepts, automatically included in the alignment results, were used to identify common concepts across the ontologies.

3.4 Concept of Correspondences

The output of this approach consisted of establishing, whenever possible, correspondences between concepts from the source and target ontologies. At this stage, the highest similarity scores obtained from the vector comparison step were identified and included as part of the alignment results. Moreover, a subset of results derived from the previous labeling step was integrated with the best computed similarities to form the final set of correspondences.

4 Experiments

Ontology alignment approaches were commonly validated using the OAEI, which provided standardized ontologies for benchmarking. In 2024, several ontology packages were made available to facilitate experimentation with alignment methods. In this work, we utilized a selection of these ontologies.

Two repositories were downloaded from <https://dwslab.github.io/melt/track-repository> to conduct the experiment. Each repository contained four XML files: parameters (specifying the entities to be matched), reference (listing the expected correspondences), source (the ontology to be aligned), and target (the ontology to which the source will be aligned). In the present approach, only concepts were considered for alignment.

The validation of an ontology alignment method involves computing evaluation metrics such as precision, recall, and F-measure [49], which were defined by Eqs. (5)–(7). TP, FP, and FN denote true positives, false positives, and false negatives, respectively.

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

$$F\text{-measure} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (7)$$

The experimental evaluation of the proposed approach was conducted under a controlled hardware and software configuration. All tests were performed on an HP EliteBook Folio 9480m laptop equipped with an Intel® Core™ i5-4210U processor (4 cores, base frequency 1.70 GHz up to 2.4 GHz in turbo mode), 16 GB of RAM, and running Windows 10 Professional 64-bit. The implementation was carried out in Python 3.11.2 with the use of several specialized libraries: Owlready2 for ontology processing and querying whereas NumPy for matrix manipulation.

4.1 Results

Table 1 presents the metrics of the ontologies adopted in this study and they were derived from the RDF triples extracted from each ontology. Case 1 corresponds to the Cmt-conference repository, while Case 2 refers to the Ce-crack-ce-crack repository.

The approach proposed in this work facilitated the discovery of correspondences beyond those explicitly referenced by the OAEI. Specifically, the similarity scores of valid correspondences ranged from 0.7 to 1. In Case 1, twenty-five correspondences were identified, of which sixteen matched entries in the reference set, while the remaining nine correspondences exhibited similarity scores below one. In Case 2, out of the fourteen identified correspondences, ten corresponded to thirteen referenced correspondences. These validation results are summarized in Table 2.

The selected ontologies (Cmt-conference and Ce-crack-ce-crack) were representative as they covered different domains and structural complexities, including both hierarchical and interconnected-concept networks. This allowed the method to be tested on varied semantic relationships. Despite using only two repositories, the HMM-based approach was expected to generalize larger or structurally different ontologies, since it modelled probabilistic dependencies among concepts rather than relying on ontology size or specific topology. This could ensure robust performance across diverse scenarios.

Table 1. Ontology metrics

Ontologies	Concepts	Relationships
Cmt-conference		
Source	29	50
Target	59	47
Source + target	82	95
Ce-crack-ce-crack		
Source	208	57
Target	485	64
Source + target	678	84

Table 2. Validation metrics of ontology alignment

Case	TP	FP	FN	Precision	Recall	F-Measure
Case 1	16	0	9	1.00	0.64	0.78
Case 2	10	3	1	0.77	0.91	0.83

Note: TP = true positives; FP = false positives; FN = false negatives.

4.2 Discussion

The alignment approach proposed in this study leveraged both the lexical and structural characteristics of the ontologies under consideration. Initially, concepts were compared at the lexical level, primarily based on the similarity of their labels or terms. This lexical comparison was subsequently complemented by an analysis of the structural organization of concepts within the ontology. The final similarity score between two concepts was determined based on the highest similarity value obtained across both levels.

For the computation of lexical similarity, the cosine similarity metric was employed. This measure was particularly well-suited for comparing concept vectors with small numerical values, as it remained sensitive to fine-grained distinctions in terms of usage and distribution. The structural component of the comparison accounted for the hierarchical position of each concept and its relationships within the ontology. Table 1 presents metrics regarding the number of concepts and their relationships extracted from the ontologies with targeted SPARQL queries. These metrics offered an indication of the complexity and semantic richness in the ontologies.

As shown in Table 2, the quality of the alignment improved significantly when ontologies contained a greater number and diversity of semantic relationships. In other words, the richer the relational structure of an ontology, the more accurate and meaningful the resulting alignments tend to be. This observation underscored the importance of well-modeled ontologies with detailed semantic networks, particularly for tasks that demanded high-precision ontology mapping.

A brief examination of alignment errors showed that false positives mainly occurred when concepts shared semantic similarity but differed syntactically, while false negatives arose from ambiguous labels or sparse structural connections. For instance, in Case 2, a source concept related to authornotreviewer was incorrectly matched to meta-reviewer, and in Case 1, several valid correspondences were missed due to limited relational context. These observations indicated that enhancing structural and semantic features, such as logical axioms or property restrictions, could further minimize alignment errors.

The proposed approach was restricted by several key limitations that warrant careful consideration. First, it did not consider logical axioms or property restrictions (e.g., owl:allValuesFrom, owl:someValuesFrom, or disjointness constraints), which might carry significant semantic and syntactic information and thus affect the depth and precision of the alignments. Second, the study could incur considerable computational cost as the number of concepts increased, since each concept was represented by a vector whose dimensionality grew alongside the ontology’s size and relational complexity. This could lead to longer pairwise comparison times. Third, the method was sensitive to sparse relational structures, where limited connections between concepts constrain the information available for

alignment, potentially resulting in false negatives or uncertain correspondences. To address these methodology challenges, future work should focus on incorporating additional semantic features, optimizing vector computation and comparison strategies to save overheads, as well as exploring hierarchical, sampling-based, or probabilistic aggregation techniques to improve scalability and robustness across large and structurally diverse ontologies.

5 Conclusions

A novel ontology alignment approach was proposed in this study and its emergence was based on machine learning, specifically utilizing HMMs. The method involves transforming each ontology into an HMM to capture its structural organization in terms of concepts and relationships. RDF triples were extracted via targeted SPARQL queries. The state transition probability matrix of each HMM was interpreted as a set of vectors denoting individual concepts; cosine similarity was applied to these vectors to assess the degree of similarity between concepts. Under this evaluation, concepts exhibiting similarity scores approaching 1 were considered aligned.

Experimental evaluation conducted on datasets provided by the OAEI demonstrated that the proposed approach effectively identified correspondences with appreciable accuracy. It should be noted that the current method aligned concepts only but did not handle axioms; consequently, its performance was influenced by the richness of the ontology in terms of relationships. Future work should address these few limitations by incorporating axioms to elevate alignment quality and exploring the contribution of extra HMM parameters.

Author Contributions

Conceptualization, D.J. and W.L.; methodology, D.J. and M.K.A.; software, D.J.; validation, W.L. and K.G.; formal analysis, M.J.; resources, W.L.; writing—original draft preparation, D.J. and W.L.; writing—review and editing, M.K.A. and K.; supervision, K.G. and K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data (ontologies) supporting our research results were deposited in <https://dwsllab.github.io/melt/track-repository>, which does not issue DOIs.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] B. Chandrasekaran, J. R. Josephson, and V. R. Benjamins, “What are ontologies, and why do we need them?” *IEEE Intell. Syst.*, vol. 14, no. 1, pp. 20–26, 1999. <http://doi.org/10.1109/5254.747902>
- [2] A. Schutz and P. Buitelaar, “Rellex: A tool for relation extraction from text in ontology extension,” in *The Semantic Web—ISWC 2005: 4th International Semantic Web Conference, ISWC 2005, Galway, Ireland, November 6–10, 2005. Proceedings 4*. Springer, 2005, pp. 593–606. http://doi.org/10.1007/11574620_43
- [3] J. Pérez, M. Arenas, and C. Gutierrez, “Semantics and complexity of SPARQL,” in *International Semantic Web Conference*. Springer, 2006, pp. 30–43. http://doi.org/10.1007/11926078_3
- [4] F. Ensan and W. Du, “A modular approach to scalable ontology development,” in *Canadian Semantic Web: Technologies and Applications*. Springer, 2010, pp. 79–103. http://doi.org/10.1007/978-1-4419-7335-1_4
- [5] Ontology Alignment Evaluation Initiative, “OAEI 2024 Campaign,” 2024. <https://oaei.ontologymatching.org/2024/>
- [6] L. Warda, S. S. Ahmed, O. Hayatou, G. Kaladzavi, and A. Samdalle, “A hidden markov model-based approach for lightweight ontology modularization using k-means clustering,” *Ing. Syst. Inf.*, vol. 29, no. 1, p. 57, 2024. <https://doi.org/10.18280/isi.290107>
- [7] B. B. Alhassan, S. B. Junaidu, and A. A. Obiniyi, “Extending an ontology alignment system with a lexical database,” *Sci. Res. J.*, vol. 3, no. 1, pp. 12–17, 2015. <https://scirj.org/rp/files/original/73339aaf530de5ba7a4ba897533cd58d.pdf>
- [8] X. Xue and X. Wu, “Optimizing biomedical ontology alignment in lexical vector space,” *J. Intell. Fuzzy Syst.*, vol. 38, no. 5, pp. 5609–5614, 2020. <http://doi.org/10.3233/JIFS-179650>
- [9] F. Q. Real, G. Bella, F. McNeill, and A. Bundy, “Using domain lexicon and grammar for ontology matching,” in *The Fifteenth International Workshop on Ontology Matching*. Virtual Conference (originally planned in Athens, Greece), 2020, pp. 1–12. <https://www.research.ed.ac.uk/en/publications/using-domain-lexicon-and-grammar-for-ontology-matching/>
- [10] A. Patel and S. Jain, “A novel approach to discover ontology alignment,” *Recent Adv. Comput. Sci. Commun.*, vol. 14, no. 1, pp. 273–281, 2021. <http://doi.org/10.2174/2666255813666191204143256>

- [11] S. Ibrahim, S. Fathalla, J. Lehmann, and H. Jabeen, “Toward the multilingual semantic web: Multilingual ontology matching and assessment,” *IEEE Access*, vol. 11, pp. 8581–8599, 2023. <http://doi.org/10.1109/ACCESS.2023.3238871>
- [12] Y. Peng, M. Alam, and T. Bonald, “Ontology matching using textual class descriptions,” in *Proceedings of the 18th International Workshop on Ontology Matching co-located with the 22nd International Semantic Web Conference (OM@ISWC 2023)*, Athens, Greece, 2023, pp. 67–72. <https://telecom-paris.hal.science/hal-04459105/>
- [13] F. Schneider, S. Dash, S. Bagchi, N. Mihindukulasooriya, and A. M. Gliozzo, “Nlfoa: Natural language focused ontology alignment,” in *Proceedings of the 12th Knowledge Capture Conference 2023*, Pensacola, FL, United States, 2023, pp. 114–121. <http://doi.org/10.1145/3587259.3627560>
- [14] S. Guo, C. Wang, Y. Chen, K. Liu, R. Li, and J. Zhao, “EventOA: An event ontology alignment benchmark based on framenet and wikidata,” in *Findings of the Association for Computational Linguistics: ACL 2023*, Toronto, ON, Canada, 2023, pp. 10 038–10 052. <https://doi.org/10.18653/v1/2023.findings-acl.637>
- [15] J. Silva, K. Revoredo, F. Araujo Baião, and C. Lima, “Enhancing ontology matching: Lexically and syntactically standardizing ontologies through customized lexical analyzers,” *Semant. Web*, vol. 16, no. 4, pp. 1–41, 2025. <https://doi.org/10.1177/22104968251344447>
- [16] A. Sharma, S. Jain, and A. Patel, “Large scale ontology matching system (LSMatch),” *Recent Adv. Comput. Sci. Commun.*, vol. 17, no. 2, pp. 20–30, 2024. <http://doi.org/10.2174/2666255816666230606140526>
- [17] S. C. Chu, X. Xue, J. Pan, and X. Wu, “Optimizing ontology alignment in vector space,” *J. Internet Technol.*, vol. 21, no. 1, pp. 15–22, 2020. <https://jit.ndhu.edu.tw/article/view/2218>
- [18] Y. Xiang, Z. Zhang, J. Chen, X. Chen, Z. Lin, and Y. Zheng, “OntoEA: Ontology-guided entity alignment via joint knowledge graph embedding,” in *Findings of the Association for Computational Linguistics: ACL 2021*, Online, 2021, pp. 1117–1128. <http://doi.org/10.18653/v1/2021.findings-acl.96>
- [19] L. Guo, Q. Zhang, Z. Sun, M. Chen, W. Hu, and H. Chen, “Understanding and improving knowledge graph embedding for entity alignment,” in *International Conference on Machine Learning*, Baltimore, MD, United States, 2022, pp. 8145–8156. <https://proceedings.mlr.press/v162/guo22i.html>
- [20] B. Hnatkowska, A. Kozierkiewicz, and M. Pietranik, “Fuzzy logic framework for ontology instance alignment,” in *International Conference on Computational Science*, Brunel University London, London, United Kingdom, 2022, pp. 653–666. https://doi.org/10.1007/978-3-031-08754-7_68
- [21] Z. Hao, W. Mayer, J. Xia, G. Li, L. Qin, and Z. Feng, “Ontology alignment with semantic and structural embeddings,” *J. Web Semant.*, vol. 78, p. 100798, 2023. <https://doi.org/10.1016/j.websem.2023.100798>
- [22] F. Şentürk and V. Aytac, “A graph-based ontology matching framework,” *New Gener. Comput.*, vol. 42, no. 1, pp. 33–51, 2024. <https://doi.org/10.1007/s00354-022-00200-3>
- [23] V. Iyer, A. Agarwal, and H. Kumar, “Veealign: A supervised deep learning approach to ontology alignment,” in *The Fifteenth International Workshop on Ontology Matching*, Virtual Conference (originally planned: Athens, Greece), 2020, pp. 216–224. <https://moex.inria.fr/files/reports/ISWC2020-OM-ws.pdf#page=225>
- [24] J. Chakraborty, S. K. Bansal, L. Virgili, K. Konar, and B. Yaman, “Ontoconnect: Unsupervised ontology alignment with recursive neural network,” in *Proceedings of the 36th Annual ACM Symposium on Applied Computing*, Virtual Event, Republic of Korea, 2021, pp. 1874–1882. <https://doi.org/10.1145/3412841.3442059>
- [25] X. Xue, C. Yang, C. Jiang, P. W. Tsai, G. Mao, and H. Zhu, “Optimizing ontology alignment through linkage learning on entity correspondences,” *Complexity*, vol. 2021, no. 1, pp. 1–12, 2021. <http://doi.org/10.1155/2021/5574732>
- [26] J. Chen, E. Jiménez-Ruiz, I. Horrocks, D. Antonyrajah, A. Hadian, and J. Lee, “Augmenting ontology alignment by semantic embedding and distant supervision,” in *The Semantic Web: 18th International Conference, ESWC 2021*. Springer, 2021, pp. 392–408. https://doi.org/10.1007/978-3-030-77385-4_23
- [27] J. Hao, C. Lei, V. Efthymiou, A. Quamar, F. Özcan, Y. Sun, and W. Wang, “Medto: Medical data to ontology matching using hybrid graph neural networks,” in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, Virtual Event, Singapore, 2021, pp. 2946–2954. <http://doi.org/10.1145/3447548.3467138>
- [28] X. Xue and Q. Huang, “Generative adversarial learning for optimizing ontology alignment,” *Expert Syst.*, vol. 40, no. 4, p. e12936, 2023. <http://doi.org/10.1111/exsy.12936>
- [29] J. Fürst, M. Fadel Argerich, and B. Cheng, “Versamatch: Ontology matching with weak supervision,” *Proc. VLDB Endow.*, vol. 16, no. 6, pp. 1305–1318, 2023. <http://doi.org/10.14778/3583140.3583148>
- [30] S. S. Norouzi, M. S. Mahdavejad, and P. Hitzler, “Conversational ontology alignment with ChatGPT,” in *Proceedings of the 18th International Workshop on Ontology Matching co-located with the 22nd International Semantic Web Conference (OM@ISWC 2023)*, Athens, Greece, 2023, pp. 61–66. <https://doi.org/10.48550/arXiv.2308.09217>

- [31] P. G. Cotovio, L. Ferraz, D. Faria, L. Balbi, M. C. Silva, and C. Pesquita, "Matcha-DL: A tool for supervised ontology alignment," *Semant. Web*, 2024, preprint. <https://www.semantic-web-journal.net/system/files/swj3648.pdf>
- [32] C. Touati and A. Kemmar, "Deep reinforcement learning approach for ontology matching problem," *Int. J. Data Sci. Anal.*, vol. 18, no. 1, pp. 97–112, 2024. <http://doi.org/10.1007/s41060-023-00425-5>
- [33] B. Sangeetha and R. Vidhyapriya, "A novel accurate and time efficient map reduce approach for biomedical ontology alignment," *J. Electr. Eng. Technol.*, vol. 19, no. 3, pp. 1761–1773, 2024. <http://doi.org/10.1007/s42835-023-01616-7>
- [34] N. Gupta, P. K. Verma, S. Raj, Anushree, N. Rakesh, and M. Gulhane, "An ontology alignment based on machine learning for integration of patient health data," *Int. J. Comput. Digit. Syst.*, vol. 16, no. 1, pp. 189–198, 2024. <https://journal.uob.edu.bh/items/b8b17a4c-a6ea-49ad-8dbe-ee25b557ddd4>
- [35] H. B. Giglou, J. D'Souza, F. Engel, and S. Auer, "LLMs4OM: Matching ontologies with large language models," in *The Semantic Web: ESWC 2024 Satellite Events*, Hersonissos, Crete, Greece, 2024, pp. 25–35. https://doi.org/10.1007/978-3-031-78952-6_3
- [36] A. Bento, A. Zouaq, and M. Gagnon, "Ontology matching using convolutional neural networks," in *Proceedings of the Twelfth Language Resources and Evaluation Conference*, Marseille, France, 2020, pp. 5648–5653. <https://aclanthology.org/2020.lrec-1.693/>
- [37] S. Neutel and M. H. de Boer, "Towards automatic ontology alignment using BERT," in *AAAI Spring Symposium: Combining Machine Learning with Knowledge Engineering*, Stanford University, Palo Alto, California, United States, 2021, pp. 1–12. <https://ceur-ws.org/Vol-2846/paper28.pdf>
- [38] Y. He, J. Chen, D. Antonyrajah, and I. Horrocks, "Biomedical ontology alignment with BERT," in *16th International Workshop on Ontology Matching (OM@ISWC 2021)*, 2021, pp. 1–12. <https://ora.ox.ac.uk/object/uuid:55395994-d156-4744-9dd6-b8b499e1db29>
- [39] Y. He, J. Chen, D. Antonyrajah, and I. Horrocks, "BERTmap: A BERT-based ontology alignment system," in *Proceedings of the 36th AAAI Conference on Artificial Intelligence (AAAI 2022)*, 2022, pp. 5684–5691. <http://doi.org/10.1609/aaai.v36i5.20510>
- [40] S. Javed, F. Sandin, H. Mokayed, J. Delsing, and M. Liwicki. (2022) Deep ontology alignment with BERT int: Improvements and industrial internet of things (IIoT) case study. <https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1651854&dsid=2772>
- [41] X. Xue, "Complex ontology alignment for autonomous systems via the compact co-evolutionary brainstorm optimization algorithm," *ISA Trans.*, vol. 132, pp. 190–198, 2023. <http://doi.org/10.1016/j.isatra.2022.05.034>
- [42] L. L. Snijder, Q. T. S. Smit, and M. H. T. de Boer, "Advancing ontology alignment in the labor market: Combining large language models with domain knowledge," in *AAAI 2024 Spring Symposium: Empowering Machine Learning and Large Language Models with Domain and Commonsense Knowledge*, Stanford, CA, United States, 2024, pp. 253–262. <http://doi.org/10.1609/aaais.v3i1.31208>
- [43] F. Hartendorp, T. Seinen, E. van Mulligen, and S. Verberne, "Biomedical entity linking for Dutch: Fine-tuning a self-alignment BERT model on an automatically generated Wikipedia corpus," in *Proceedings of the First Workshop on Patient-Oriented Language Processing (CL4Health) @ LREC-COLING 2024*, Torino, Italia, 2024, pp. 253–263. <https://aclanthology.org/2024.cl4health-1.31/>
- [44] R. Amini, S. S. Norouzi, P. Hitzler, and R. Amini, "Towards complex ontology alignment using large language models," in *Knowledge Graphs and Semantic Web: 6th International Conference, KGSWC 2024*, Paris, France, 2024, pp. 17–31. https://doi.org/10.1007/978-3-031-81221-7_2
- [45] A. Slimane, "Techniques et outils pour l'alignement d'ontologies," masterthesis, Université Mouloud Mammeri de Tizi-Ouzou, 2017. <https://dspace.ummto.dz/server/api/core/bitstreams/24938f4a-e5ee-42ec-9e88-cecfba4e418c/content>
- [46] L. Warda, G. Kaladzavi, and A. Samdalle, "Integration of ontology transformation into hidden markov model," *Inf. Dyn. Appl.*, vol. 1, no. 1, pp. 2–13, 2022. <http://doi.org/10.56578/ida010102>
- [47] D. Panneer, K. Ragunathan, M. Ramalingam, and L. K. Narayanan, "Comparative study on ontology matching tools and methods," in *AIP Conference Proceedings*, Chennai, India, 2022, p. 030002. <http://doi.org/10.1063/5.0188047>
- [48] I. Osman, S. B. Yahia, and G. Diallo, "Ontology integration: Approaches and challenging issues," *Inf. Fusion*, vol. 71, pp. 38–63, 2021. <http://doi.org/10.1016/j.inffus.2021.01.007>
- [49] H. Khan, M. Saqib, H. A. Khattak, S. I. Ali, and S. Lee, "Ontology alignment for accurate ontology matching: A survey," in *Digital Health Transformation, Smart Ageing, and Managing Disability*. Springer, 2023. https://doi.org/10.1007/978-3-031-43950-6_31